LIATE

1. $\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x = e^x(x-1) + C$

$u = x \rightarrow du = dx$
$dv = e^x dx \rightarrow v = e^x$

2. $\int x^2 e^{3x}\,dx = x^2\cdot\frac{e^{3x}}{3} - \int \frac{e^3}{3}\cdot 2x\,dx = \frac{1}{3}x^2\cdot e^{3x} - \frac{1}{3}\int e^{3x}2x\,dx$

$u = x^2 \rightarrow du = 2x\,dx$
$dv = e^{3x}dx \rightarrow v = \frac{e^{3x}}{3}$

$= \frac{1}{3}x^2e^{3x} - \frac{1}{3}\left(\frac{2xe^{3x}}{3} - \int \frac{e^{3x}}{3}2\,dx\right)$

$u = 2x \rightarrow du = 2dx$
$dv = e^{3x}dx \quad v = \frac{e^{3x}}{3}$

$= \frac{1}{3}x^2e^{3x} - \frac{2}{9}xe^{3x} + \frac{2}{9}\int e^{3x}dx = \frac{1}{3}x^2e^{3x} - \frac{2}{9}xe^{3x} + \frac{2e^{3x}}{27} + C$

3. $\int xe^{-3x}\,dx = -\frac{xe^{-3x}}{3} - \int -\frac{e^{-3x}}{3}\,dx = -\frac{xe^{-3x}}{3} + \frac{1}{3}\int e^{-3x}dx = -\frac{xe^{-3x}}{3} + -\frac{e^{3x}}{9} + C$

$u = x \rightarrow du = dx$
$dv = e^{-3x}dx \rightarrow v = -\frac{e^{-3x}}{3}$

4. $\int \ln(x)\,dx = x\ln(x) - \int dx = x\ln(x) - x + C$

$u = \ln(x) \rightarrow du = \frac{1}{x}dx$
$dv = dx \rightarrow v = x$

5. $\int x\,sen(3x)\,dx = -x\cos(3x) + \int \cos(3x)\,dx = -x\cos(3x) + \sin(3x) + C$

$u = x \rightarrow du = dx$
$dv = sen(3x)dx \rightarrow v = -\cos(3x)$

6. $\int x\sec^2(x)\,dx = x\tan(x) - \int \tan(x)\,dx = x\tan(x) + \ln|\cos(x)| + C$

$u = x \rightarrow du = dx$
$dv = \sec^2(x)dx \quad v = \tan(x)$

7. $\int x\arctan(x)\,dx$

8. $\int e^x \cos(x)dx = e^x\cos(x) + \int e^x \sin(x)\,dx$

$\begin{cases} u = \sin(x) \rightarrow du = \cos(x)dx \\ dv = e^x dx \rightarrow v = e^x \end{cases}$

cíclicas

$u = \cos(x) \rightarrow du = -\sin(x)dx$
$dv = e^x dx \rightarrow v = e^x$

$= e^x\cos(x) + e^x\sin(x) - \int e^x\cos(x)\,dx$

$= \dfrac{e^x\cos(x) + e^x\sin(x)}{2} + C$

9. $\int (x^2-3x+5)e^x dx = (x^2-3x+5)e^x - \int e^x(2x-3)\,dx$

$u = x^2-3x+5 \rightarrow du = 2x-3\,dx$
$dv = e^x dx \rightarrow v = e^x$

$= e^x(x^2-3x+5) + (2x-3)e^x - \int 2e^x dx$

$= e^x(x^2-3x+5) + e^x(2x-3) - 2e^x + C$

10. $\int x^5 \ln(x)dx = \dfrac{x^6}{6}\ln(x) - \dfrac{1}{6}\int \dfrac{x^6}{x}dx = \dfrac{x^6}{6}\ln(x) - \dfrac{1}{6}\int x^5 dx$

$u = \ln(x) \rightarrow du = \dfrac{1}{x}dx$
$dv = x^5 dx \rightarrow v = \dfrac{x^6}{6}$

$= \dfrac{x^6}{6}\ln(x) - \dfrac{x^6}{36} + C$

11. $\int x^2 \ln(x^{\frac{1}{3}})dx = \ln(x^{1/3})\dfrac{x^3}{3} - \dfrac{1}{3}\int \dfrac{x^3}{x^{1/3}}dx = \ln(x^{1/3})\dfrac{x^3}{3} - \dfrac{1}{3}\int x^{8/3}dx = \ln(x^{1/3})\dfrac{x^3}{3} - \dfrac{x^{11/3}}{11} + C$

$u = \ln(x^{1/3}) \rightarrow du = \dfrac{1}{\sqrt[3]{x}}dx$
$dv = x^2 dx \rightarrow v = \dfrac{x^3}{3}$

12. $\int (\ln(x))^2 dx = (\ln(x))^2 \cdot x - 2\int \ln(x)\,dx$

$= x(\ln(x))^2 - 2(x\ln(x) - x) + C$

$u = (\ln(x))^2 \rightarrow du = \dfrac{2\ln(x)}{x}dx$
$dv = dx \rightarrow v = x$

13. $\int x 2^x \, dx = \frac{x2^x}{\ln(2)} - \int \frac{2^x}{\ln(2)} dx$ ~~$= \frac{x2^x}{\ln(2)} + \ln(2)^{-1} \frac{2^x}{\ln(2)} + \int \frac{2^x}{\ln(2)} \frac{\ln(2)^{-2}}{2} dx$~~

$u = x \rightarrow du = dx$

$dv = 2^x dx \rightarrow v = \frac{2^x}{\ln(2)}$

~~$u = (\ln(2))^{-1}$ $\quad du = -\frac{(\ln(2))^{-2}}{2}$~~

~~$dv = 2^x dx$ $\quad v = \frac{2^x}{\ln(2)}$~~

~~$du = \frac{-(\ln(2))^{-2}}{2}$~~

$= \frac{x2^x}{\ln(2)} - \int \frac{2^x}{\ln(2)} dx = \frac{x2^x}{\ln(2)} - \frac{1}{\ln(2)} \int 2^x dx = \frac{1}{\ln(2)} \left(x2^x - \frac{2^x}{\ln(2)} \right)$

$2^x (\ln(2))^{-1}$

14. $\int x \cos(5x) \, dx = x\sin(5x) - \int \sin(5x) \, dx =$ ~~$x\sin(5x) + \cos(5x)$~~

$= x\sin(5x) + \frac{1}{5}\cos(5x) + C$

$u = x \rightarrow du = dx$

$dv = \cos(5x) dx \rightarrow v = \sin(5x)$

$u = 5x$

$du = 5dx$

$\frac{du}{5} = dx$

$= x\sin(5x) + \frac{1}{5} \int \sin(u) \cdot du$

$= x\sin(5x) + \frac{1}{5}\cos(u)$

$= x\sin(5x) + \frac{1}{5}\cos(5x) + C$

15. $\int e^{2x} \cos(5x) \, dx = \frac{e^{2x}}{2} \cdot \cos(5x) + \frac{5}{2} \int e^{2x} \sin(5x) \, dx$

$= \frac{1}{2} e^{2x} \cos(5x) + \frac{5}{2} \left(\frac{1}{2} e^{2x} \sin(5x) - \frac{5}{2} \int e^{2x} \cos(5x) \, dx \right)$

$u = \cos(5x) \rightarrow du = 5\sin(5x)$

$dv = e^{2x} \rightarrow \frac{e^{2x}}{2} = v$

$u = \sin(5x) \Rightarrow 5\cos(5x)dx = du$

$dv = e^{2x} dx \Rightarrow v = \frac{e^{2x}}{2}$

$= \frac{1}{2} e^{2x} \cos(5x) + \frac{5}{4} e^{2x} \sin(5x) - \frac{25}{4} \int e^{2x} \cos(5x) \, dx$

$\frac{29}{4} \int e^{2x} \cos(5x) \, dx = \frac{1}{2} e^{2x} \cos(5x) + \frac{5}{4} e^{2x} \sin(5x) \quad \cdot \frac{4}{29}$

$= \frac{2}{29} e^{2x} \cos(5x) + \frac{5}{29} e^{2x} \sin(5x) + C$